CRM – UNIT TEST REPORT

Customer Relationship Management System

Project Name	Customer Relationship Management System (CRM)	
Testing Period	2 November 2026 – 6 November 2026	
Technology Stack	React, TypeScript, Node.js, Express.js, MongoDB	

1. Introduction

Unit testing is performed to verify individual components, functions, and modules of the CRM application. The objective is to check that each part behaves as expected when tested independently, before validating the complete application.

2. Testing Objectives

- Verify individual module functionality.
- Validate input fields and error handling.
- Check database operations.
- Verify authentication and authorization logic.
- Identify errors in individual functions and components.

3. Modules Selected for Testing

Module	**Testing Performed**
Authentication	Login validation and password verification
User Management	User creation and role validation
Lead Management	Lead creation and status updates
Customer Management	Customer creation and input validation
Interactions	Interaction creation and retrieval
Follow-ups	Follow-up scheduling and updates
Feedback	Feedback submission and retrieval
Sales	Sales record creation and updates
Loyalty	Points calculation and redemption
Retention	Retention activity management
Reports	Report data retrieval
Audit Logs	Recording user activities

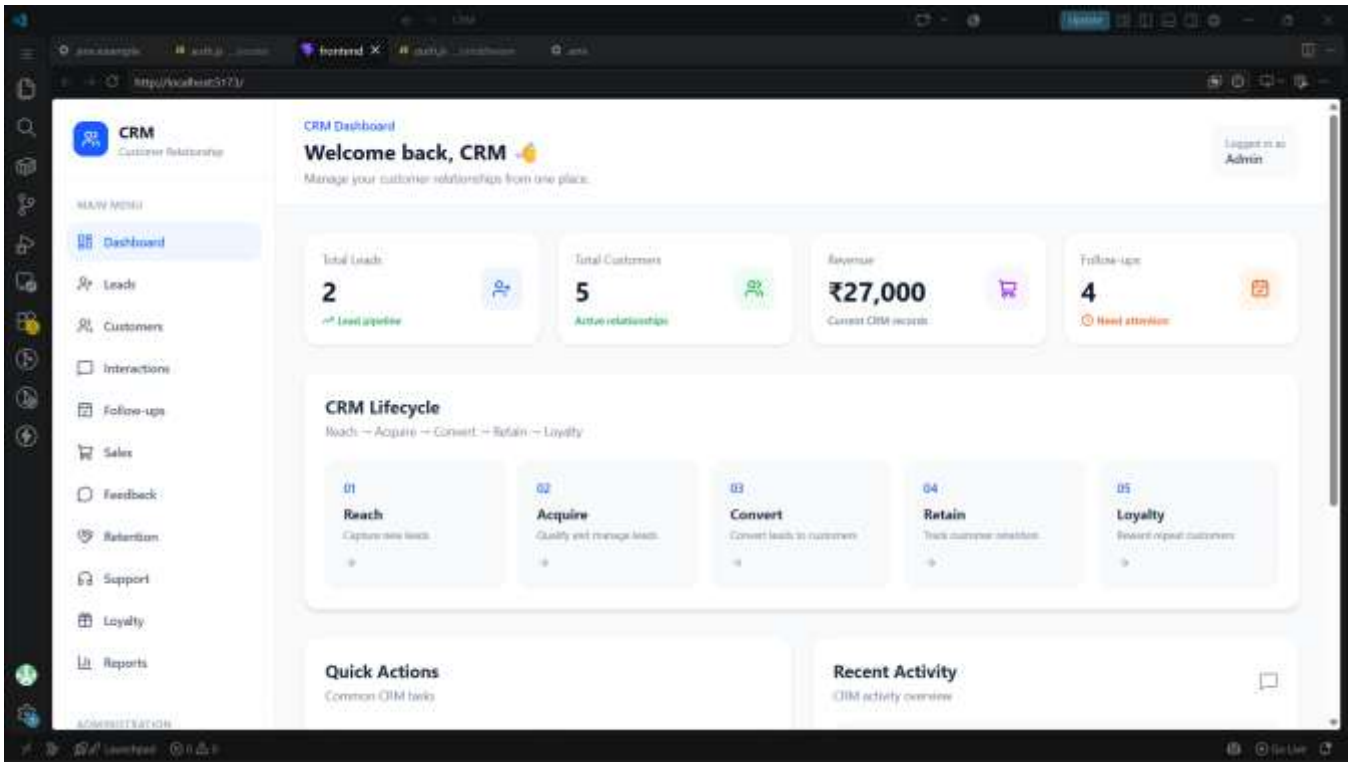
4. Test Cases

The following test cases represent the unit-level checks planned for the CRM modules. Update the execution result after performing each test.

Test ID	Test Scenario	Expected Result
UT01	Login with valid credentials	User authenticated successfully
UT02	Login with incorrect password	Error message displayed
UT03	Create a customer with valid details	Customer saved successfully
UT04	Enter an invalid phone number	Validation error displayed
UT05	Create a new lead	Lead saved successfully
UT06	Schedule a follow-up	Follow-up saved successfully
UT07	Submit customer feedback	Feedback saved successfully
UT08	Create a sales record	Sales record saved successfully
UT09	Redeem loyalty points	Points updated correctly
UT10	Record a user activity	Audit log generated

5. Screenshot Evidence

Figure 1 – Authentication Testing



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Figure 2 – Customer Module Testing

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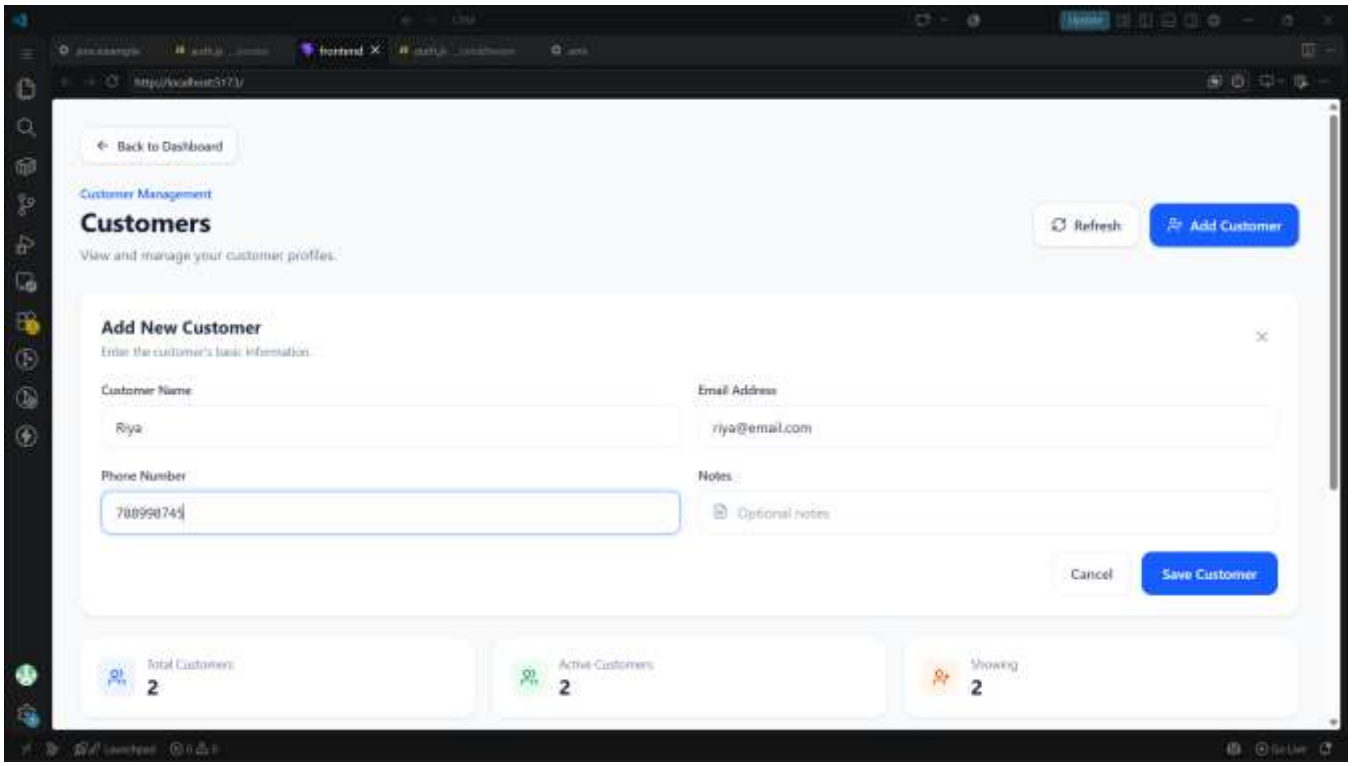


Figure 3 – Lead Management Testing

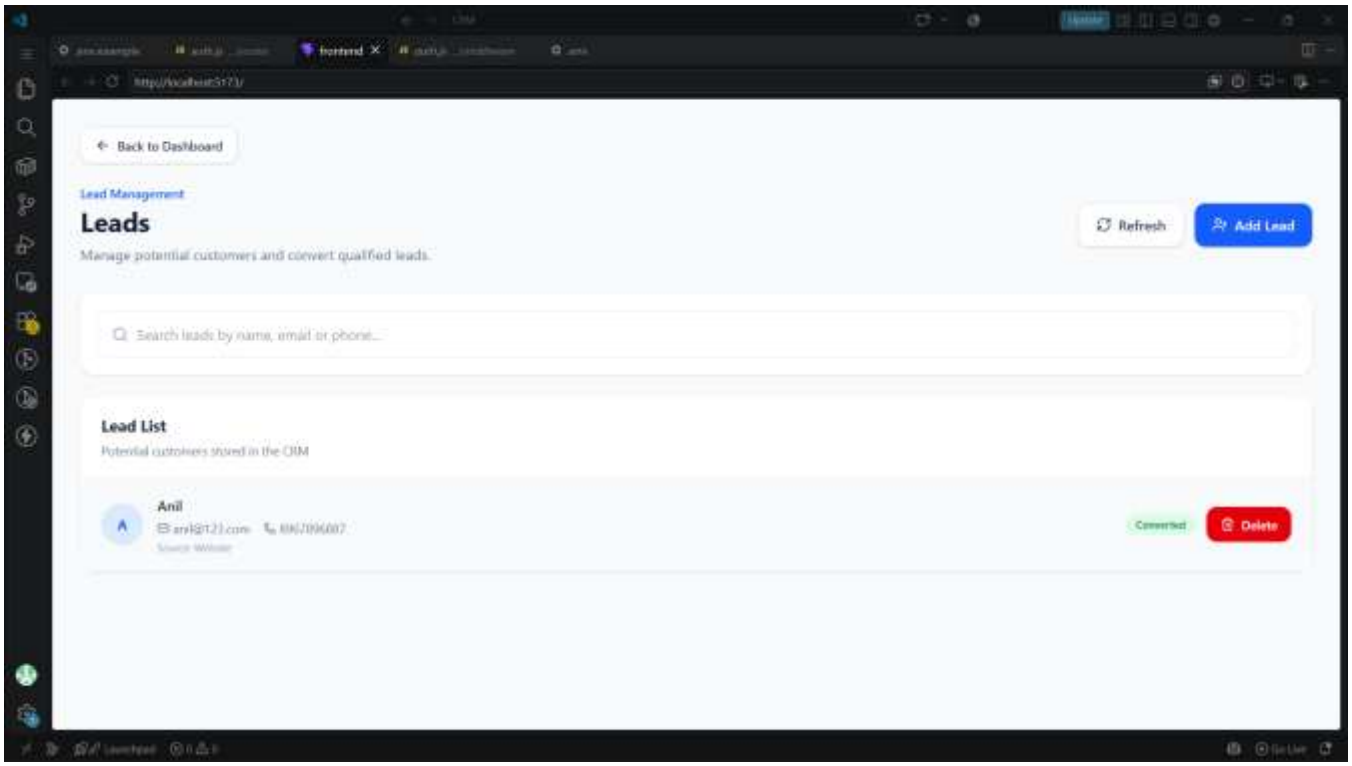


Figure 4 – Audit Log

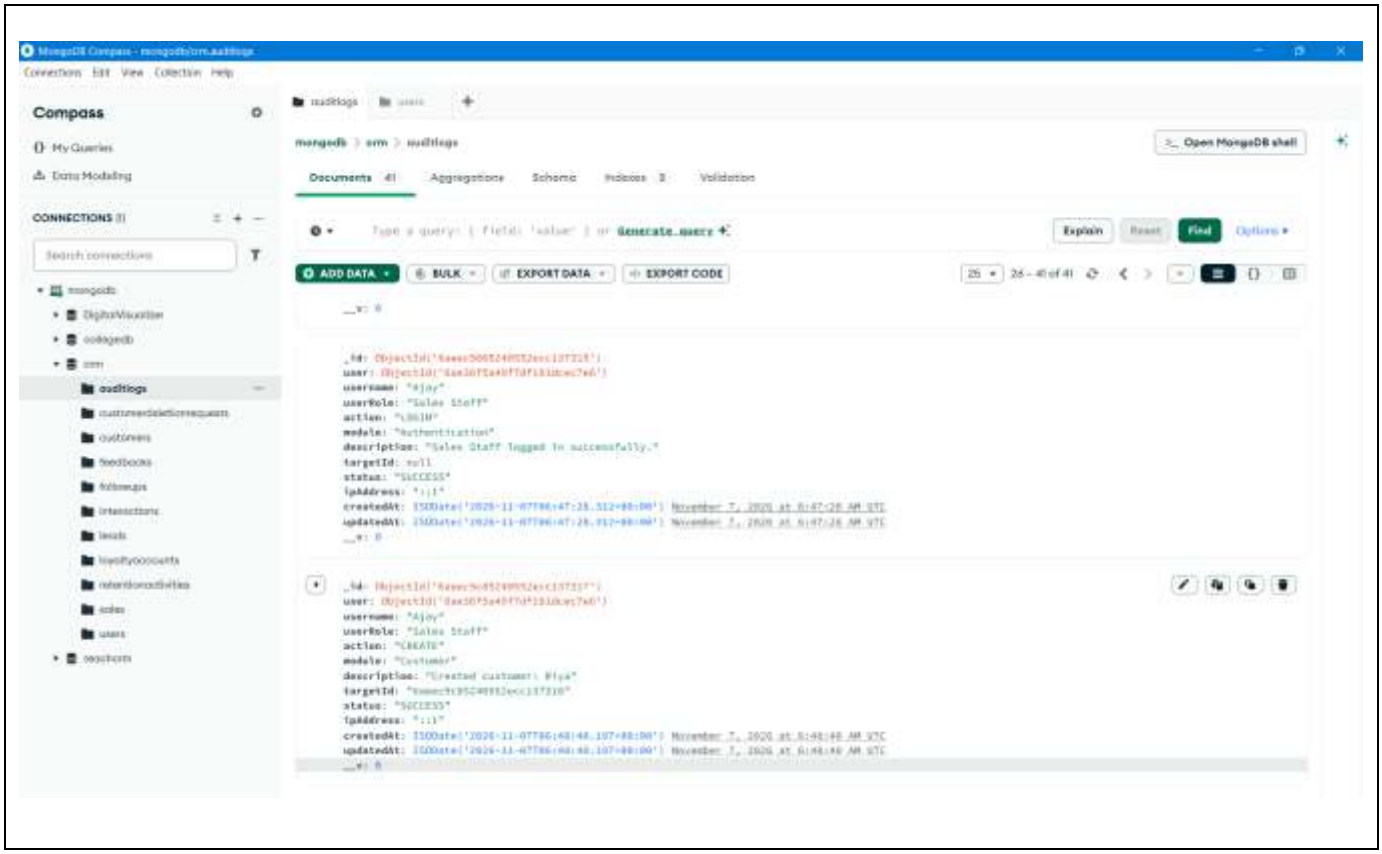


Figure 5 – Interaction

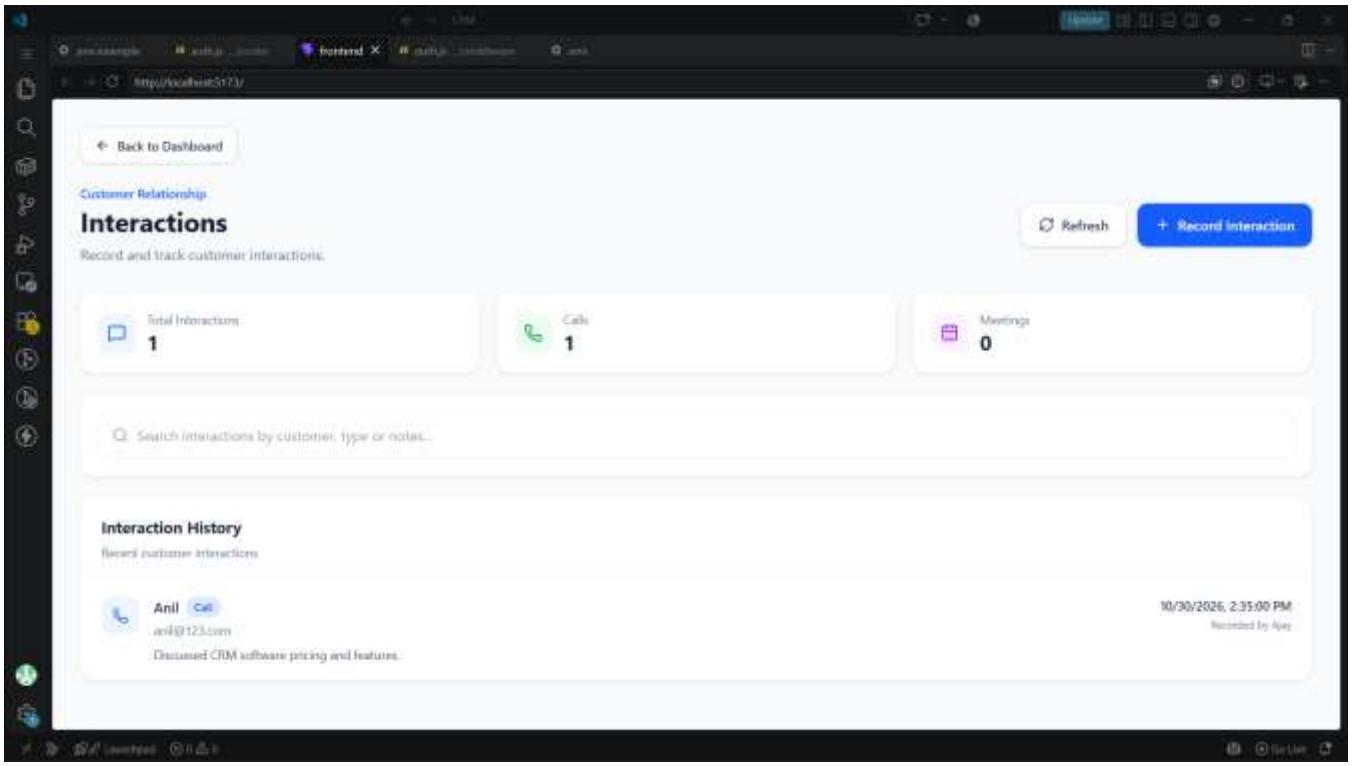


Figure 6– Feedback

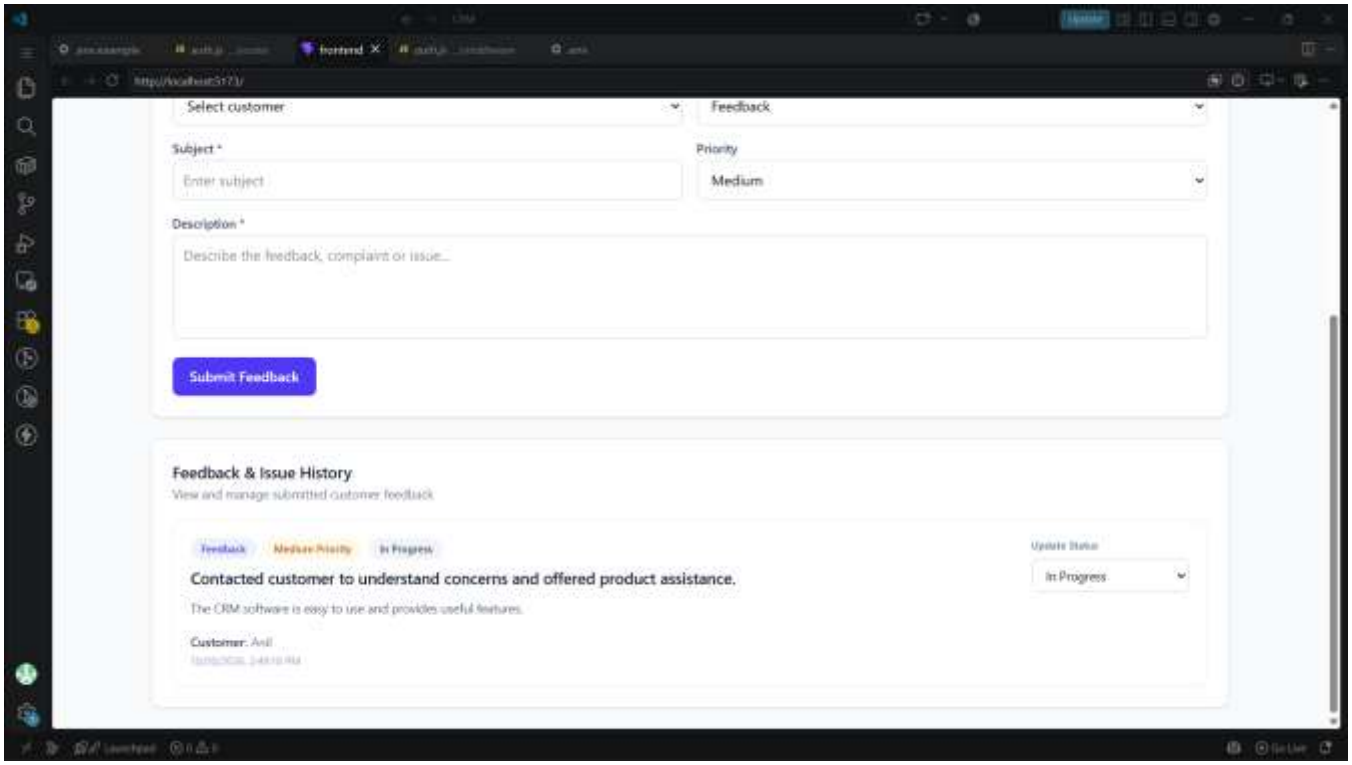


Figure 7– Sales

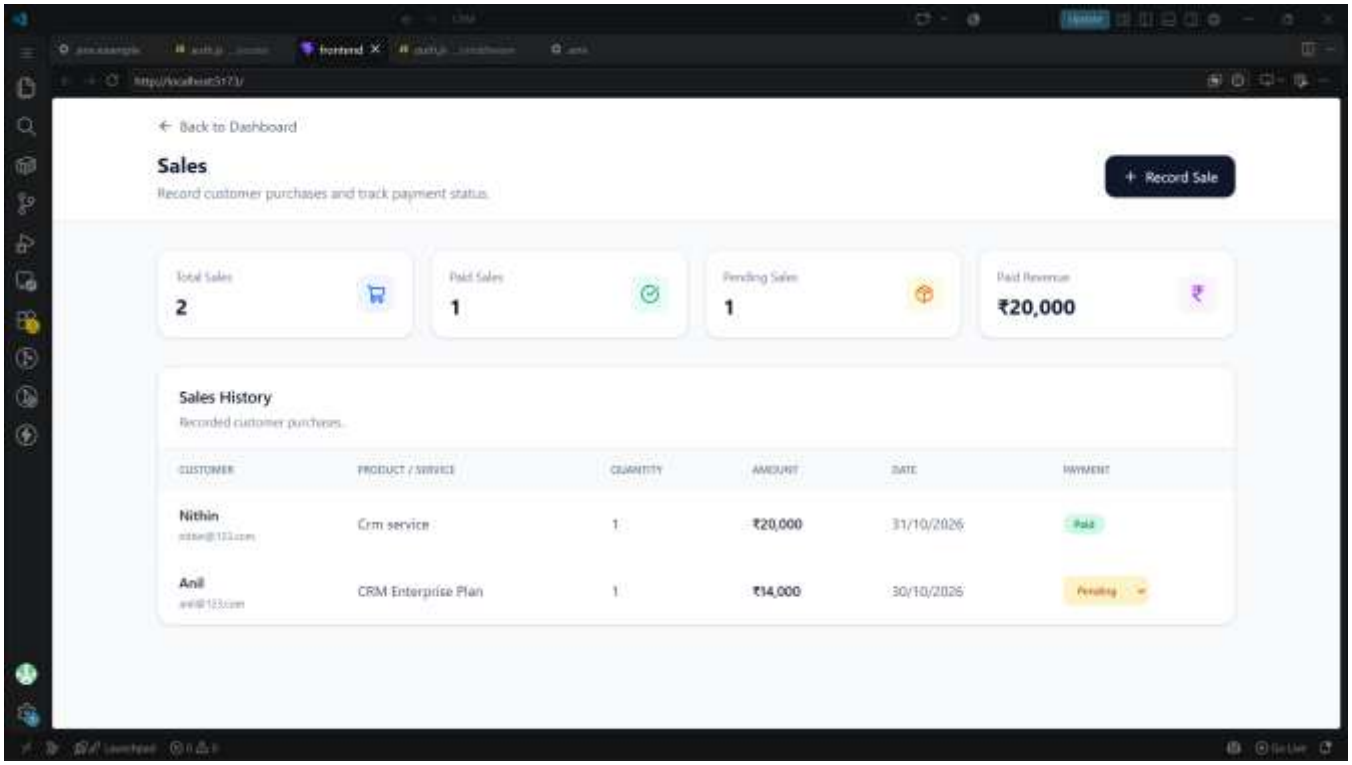
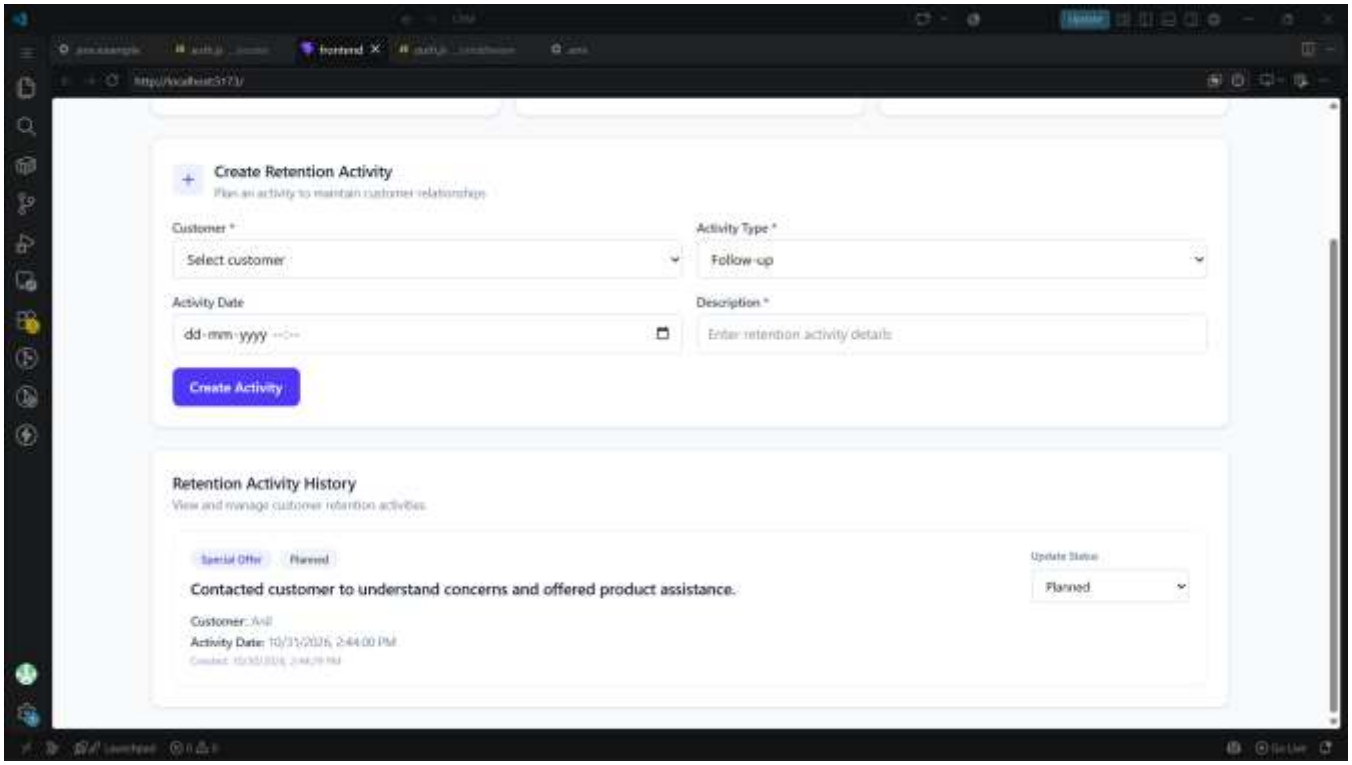


Figure 8– Retention



6. Conclusion

Unit testing helps verify the correctness of individual CRM modules. The testing focuses on authentication, data validation, database operations, and module-level functionality. Issues identified during testing should be corrected and retested before proceeding to complete system validation.